University of St.Gallen

Using the HSG Beamer template

Slide types, figures, tables, CLI agents, Overleaf

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Agenda

1. Setup: three commands to a finished deck
2. Slide types: every layout the master defines
3. Figures and tables: real research content
4. Workflows: CLI agents and Overleaf
5. Tips: golden rules and common errors

Three commands to a finished deck

1. Bootstrap a new deck folder:

```
bin/new-deck.sh ~/path/to/my-talk
```

2. Edit `deck.tex`: replace every `TODO`.
3. Compile (3 passes for remember-picture overlays to settle):

```
pdflatex deck.tex
pdflatex deck.tex
pdflatex deck.tex
```

Or one shot with `latexmk deck.tex` (uses the bundled `latexmkrc`).

Folder layout produced by `new-deck.sh`

```
my-talk/
  deck.tex                    <- you edit this
  beamerthemeHSG.sty          <- theme (do not edit)
  latexmkrc                   <- compile config
  assets/                     <- every image asset
    hsg-logo-en.pdf, hsg-badge.png, ...
  fonts/                      <- vendored Gillius ADF (xelatex/lualatex)
    GilliusADF-Regular.otf, ...
```

The theme adds `\graphicspath{{./assets/}{./}}` so `\includegraphics{hsg-logo-en.pdf}` resolves whether assets are in `./assets/` or flat next to the deck.

Preamble: set once

```latex
\documentclass[aspectratio=169]{beamer}
\usetheme{HSG}

\title{My talk}
\subtitle{Optional second line}
\author{Prof. Dr. Author Name}
\institute{Your Institute\\University of St.\,Gallen}
\date{}
\coursefooter{Course or project}
\coverimage{cover-photo.jpg}      % optional hero image

\hsgclosingcontact
  {Prof. Dr. Author Name}
  {author@unisg.ch}

\hsgsetsections{Intro, Methods, Results, Outlook}
```

The cover slide *(Titel Slide 3)*

```
\begin{frame}[plain]\titlepage\end{frame}
```

Just one line in the body. The cover photo comes from `\coverimage{...}` in the preamble. The HSG green badge is layered behind the photo (the badge corners peek through where the photo's clip cuts off).

Tip: the cover photo's right edge is clipped with a slanted polygon. Pick a photo whose subject sits in the *left* third of the frame so the clip does not crop it.

Agenda *(Agenda layout)*

```
\hsgagenda{
  \item Introduction
  \item Methodology
  \item Results
  \item Discussion
  \item Conclusion
}
```

Use 4–6 items, 1–2 words each. The same items should appear in `\hsgsetsections` so the sidebar progress matches.

Section advance *(silent, no slide rendered)*

```
\hsgsection{Methods}      % advances the sidebar
% next slide opens "Methods" with a green sidebar marker
```

`\hsgsection` does NOT render a divider slide. It only updates the left-margin progress strip. Place it immediately before the first content slide of each section.

Standard content slide *(Titel und Content)*

```
\begin{frame}
\frametitle{Results}
\framesubtitle{Detection performance}
\begin{itemize}
\item \hsgemph{AUC = 0.89} on the held-out split
\item False-positive rate held at 2%
\end{itemize}
\end{frame}
```

`\framesubtitle` renders as a tiny grey breadcrumb above the title (keep it under 6 words).

Two columns *(Titel zwei Content)*

Use the `\hsgtwocol` helper. It aligns the left edge of the columns with the frametitle, unlike Beamer's native `columns` which shifts content into the slide margin.

```
\hsgtwocol{%
  \textbf{Symmetric}
  \begin{itemize}\item Shared key\item Fast\end{itemize}
}{%
  \textbf{Asymmetric}
  \begin{itemize}\item Key pair\item Slow\end{itemize}
}
```

Rule: do NOT colour column headers HSGgreen. Reserve green for inline `\hsgemph`.

Two columns rendered

Symmetric

- Shared key
- Fast

Asymmetric

- Key pair
- Slow

Three columns rendered

Symmetric

- Shared key
- Fast

Asymmetric

- Key pair
- Slow

Hybrid

- Best of both
- Practical default

Three-column layout

```
\begin{columns}[T]
  \column{0.33\textwidth} ...
  \column{0.33\textwidth} ...
  \column{0.33\textwidth} ...
\end{columns}
```

Asymmetric splits work too: `0.6\textwidth` + `0.4\textwidth` for a wide-text + narrow-figure pattern.

Three-column rendered

Confidentiality

- Encryption hides content
- Adversary sees only ciphertext

Integrity

- MAC detects tampering
- Hash chains for logs

Availability

- Replication, redundancy
- Rate-limiting against DoS

Three balanced columns with `[T]` top-alignment. The CIA triad fits naturally into this layout.

Statement *(Statement 1/2)*

```
\hsgstatement{Security is a process, not a product.}
```

Renders at 36 pt HSG green bold on its own slide. Keeps the standard footer chrome. Use sparingly.

Security is a process, not a product.

Quote slides *(Zitat schwarze / weisse Schrift)*

Black-on-white:

```
\hsgquote{The cloud is just somebody else's computer.}{Anonymous}
```

White-on-green:

```
\hsgquoteinverted{The best way to predict the future is to invent it.}{Alan
    Kay}
```

"

The cloud is just somebody else's computer.

Anonymous

"

The best way to predict the future is to invent it.

Alan Kay

"

Imagination is more important than knowledge.

Albert Einstein

Full-bleed image *(Bild Vollfläche)*

```
\hsgfullimage{path/to/photo.jpg}
```

Image fills the whole 16×9 canvas; small HSG logo overlays bottom-left. Pre-crop the image to 16:9 to avoid letterboxing.

Including a figure

```
\begin{frame}
\frametitle{Campus reflecting pool}
\begin{center}
\includegraphics[width=0.7\linewidth, height=4.6cm,
keepaspectratio]{hsg-closing-campus.jpg}
\end{center}
\end{frame}
```

Always set *both* `width` and `height` so the figure fits in the content area without overflowing the footer.

Figure rendered

Caption text sits in the body, not in a `figure` float. Keep it to one or two lines.

Two figures side by side

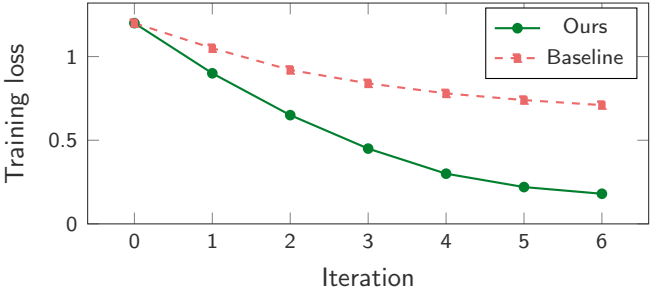
```
\begin{columns}[T]
\column{0.48\textwidth}
\includegraphics[width=\linewidth]{before.jpg}
\scriptsize\itshape Baseline.
\column{0.48\textwidth}
\includegraphics[width=\linewidth]{after.jpg}
\scriptsize\itshape Proposed.
\end{columns}
```

This covers the master's *Titel und zwei Bilder* layout.

Embedding a chart with PGFplots

```
\usepackage{pgfplots}
\pgfplotsset{compat=1.17}
\begin{tikzpicture}
\begin{axis}[width=8cm, height=4.5cm, ymin=0,
xlabel=Iteration, ylabel=Loss]
\addplot[HSGgreen, thick] coordinates {
(0,1.2) (1,0.9) (2,0.65) (3,0.45) (4,0.30)};
\end{axis}
\end{tikzpicture}
```

Chart rendered



Tables (booktabs)

```
\usepackage{booktabs}
\begin{tabular}{@{}lrr@{}}
\toprule
Method & AUC & Latency \\
\midrule
Ours & 0.89 & 12 ms \\
Baseline & 0.71 & 8 ms \\
\bottomrule
\end{tabular}
```

Table rendered

Method	AUC	Latency	Params
Ours	0.89	12 ms	4.1 M
Baseline (CNN)	0.71	8 ms	2.3 M
Baseline (LSTM)	0.74	22 ms	3.7 M
Random forest	0.62	3 ms	–

Use `booktabs` (`\toprule`, `\midrule`, `\bottomrule`) for tables. No vertical lines, no `\hline`.

CLI agent workflow

An agent (Claude Code, Codex, Cursor, Aider) scaffolds a deck cold:

```
git clone https://github.com/brunobastosrodrigues/beamer-hsg.git
cd beamer-hsg
bin/new-deck.sh ~/talks/2026-conference
cd ~/talks/2026-conference
# fill in TODO markers in deck.tex
pdflatex deck.tex && pdflatex deck.tex && pdflatex deck.tex
```

Three files the agent reads first:

- `example/skeleton.tex` – canonical starter
- `helpers.json` – machine-readable API manifest
- `AGENTS.md` – decision table + golden rules

What helpers.json gives an agent

```
{
  "frame_helpers": [
    {"name": "\\hsgstatement",
     "args": ["text"],
     "layout": "Statement 1",
     "purpose": "One bold green sentence at 36 pt."},
    ...
  ],
  "rules_to_follow": [
    "Three pdflatex passes...",
    "HSG green is reserved for...",
    ...
  ]
}
```

Every public macro: name, arity, master-layout, one-line purpose.

Overleaf workflow

Repo is private, so use the release-ZIP route (no paid Overleaf plan needed):

1. GitHub repo → *Releases* → download *Source code (zip)* for the latest tag.
2. Overleaf → *New Project* → *Upload Project* → drop the ZIP.
3. *Project Settings* → *Compiler*:
   - **XeLaTeX** for the Gill-Sans-Nova look (vendored Gillius ADF picked up automatically).
   - **pdfLaTeX** for the Lato fallback.
4. Open `example/skeleton.tex`, set as *Main document*, replace TODOs, recompile.

Full instructions: `OVERLEAF.md` in the repo root.

Golden rules

- **Never use HSG green for titles or headings.** Reserved for the sidebar, the page badge, and `\hsgemph{...}`.
- **Three passes** of `pdflatex`. Two is not enough; remember-picture overlays drift on pass two.
- **Copy ALL assets** (or use `bin/new-deck.sh`, which does it for you).
- **Keep slides spare.** If you find yourself `shrink`ing to fit, split the slide instead.
- **Cover and closing** are hand-tuned. Do not redefine their templates.

Common errors

Symptom	Where to look
Title chrome misplaced	Compile two more times – remember-picture needs three passes.
Logo or photo missing	An asset under `assets/` is missing.
Bullets render green	You added a `\setbeamercolor` in your preamble. Remove it.
Sidebar progress empty	You did not call `\hsgsetsections` in the preamble.
Page number clipped	4-digit numbers (1000+) do not fit. Split the deck.

Where to look next

- `AGENTS.md` – exhaustive agent reference + decision table from "what kind of slide do you want?" to "which helper to call".
- `example/skeleton.tex` – minimal compileable starter for any new deck.
- `examples/example.tex` – 9-slide showcase with every helper used at least once.
- `OVERLEAF.md` – step-by-step Overleaf upload.
- `helpers.json` – machine-readable API manifest.

Issues, suggestions, fixes: **github.com/brunobastosrodrigues/beamer-hsg**.

\hsgtwocol — side-by-side content

Left column

A symmetric two-column body with top alignment and equal gutters. Use it for compare/contrast, figure-plus-caption, or before/after layouts without writing a `minipage` by hand.

Right column

- Top-aligned by default
- 0.48 / 0.48 width split
- Drop *any* content inside

\hsgbibliography — references slide

Place `\hsgbibliography` just before `\hsgclosing`. Requires biblatex configured in the preamble:

```
\usepackage[backend=biber, style=numeric]{biblatex}
\addbibresource{refs.bib}
```

Long lists overflow into extra slides via `allowframebreaks`.

\hsgvideolink — clickable video thumbnail

Drop a 16:9 screenshot next to your `.tex` and call `\hsgvideolink{url}{thumbnail.jpg}`. The whole figure becomes a hyperlink that opens in the audience's browser.

Animations: \pause and \onslide

Use Beamer's standard overlay primitives. Each \pause step produces a new PDF page, so the page badge increments between clicks — same as any Beamer theme.

- First, set the scene.

Animations: \pause and \onslide

Use Beamer's standard overlay primitives. Each \pause step produces a new PDF page, so the page badge increments between clicks — same as any Beamer theme.

- First, set the scene.
- Then, motivate the gap.

Animations: \pause and \onslide

Use Beamer's standard overlay primitives. Each `\pause` step produces a new PDF page, so the page badge increments between clicks — same as any Beamer theme.

- First, set the scene.
- Then, motivate the gap.
- Finally, the contribution.

For a printable handout, add `handout` to the document class:
`\documentclass[aspectratio=169,handout]{beamer}`
Beamer collapses every overlay into a single page automatically.

Backup

Slides for questions and reference — not part of the main narrative.

Backup — Q&A material

Slides after `\hsgbackup` keep counting toward the page badge but stop inflating the section progress in the top-right corner — it shows *appendix* instead of an extra section number.

Backup — detailed numbers

Use this region for slides you want available during Q&A but not as part of the main narrative.

- Detailed derivations
- Extra results tables
- Limitations or caveats

Questions?

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"From insight to impact"